MAT 201 - MATHEMATICAL METHODS I

LECTURE 2: REVIEW OF DIFFERENTIATION AND INTEGRATION

LECTURE C24-1-2 :: TUE., NOV. 26, 2024 8:00 AM (WAT) - FRI., NOV. 29, 2024 6:00 PM (WAT)

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OUTLINE

- Introduction
- Differentiation Basics
- Integration Basics
- Derivatives of Common Functions
- Integrals of Common Functions
- Applications of Differentiation and Integration
- Conclusion

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INTRODUCTION: *DEFINITION*

- Differentiation and Integration are fundamental concepts in calculus.
- Introduced by Sir Isaac Newton and Gottfried Wilhelm Leibniz in the 17th century.
- These concepts form the basis of calculus, a branch of mathematics focused on rates of change and accumulation.

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DIFFERENTIATION AND INTEGRATION OVERVIEW

- **Differentiation**
  - Measures the rate at which a quantity changes.
  - Represents the slope of a curve at a given point.
  - Denoted by $\frac{df(x)}{dx}$ or $f'(x)$.
- **Integration**
  - Involves finding the accumulation of quantities.
  - Represents the area under a curve.
  - Denoted by $\int f(x)\,dx$.

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IMPORTANCE IN MATHEMATICS AND REAL-WORLD APPLICATIONS

- **Mathematics:** Essential tools for solving complex problems in calculus, physics, engineering, and more.
- **Physics:** Used to model and analyze physical phenomena, such as motion, electricity, and heat transfer.
- **Engineering:** Crucial in designing and optimizing systems, structures, and processes.
- **Economics:** Applied in modeling economic trends, optimization problems, and financial analysis.
- **Biology and Medicine:** Utilized for modeling biological processes, population dynamics, and medical research.

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<u>DIFFERENTIATION BASICS:</u> DEFINITION OF DERIVATIVE

- The derivative of a function measures how the output changes with respect to changes in the input.
- The derivative of a function $f(x)$ at a point x is denoted as $f'(x)$.
- Mathematically, the derivative is defined as:
  $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

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DIFFERENTIATION BASICS: NOTATION

- Prime Notation: $f'(x)$ is a common way to denote the derivative of $f(x)$.
- Limit Notation: The derivative can also be expressed using limits:
  $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$
- Example: If $f(x) = x^2$, then $f'(x) = 2x$.

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<u>DIFFERENTIATION BASICS:</u> RULES OF DIFFERENTIATION

- Constant Rule: $[c]' = 0$ for any constant c.
- Power Rule: $[x^n]' = nx^{n-1}$ for any real number n.
- Sum/Difference Rule: $[f(x) \pm g(x)]' = f'(x) \pm g'(x)$.
- Product Rule: $[f(x) \cdot g(x)]' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$.
- Quotient Rule: $\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$, for $g(x) \neq 0$.

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<u>DIFFERENTIATION BASICS:</u> EXAMPLE PROBLEMS

- Find the derivative of $f(x) = 3x^2 - 2x + 1$.
- Calculate $h'(x)$ for $h(x) = \frac{x^2+2x}{x+1}$.
- If $g(x) = e^x \sin(x)$, determine $g'(x)$.
- Solve for $f'(x)$ if $f(x) = \sqrt{x} + \frac{1}{x^2}$.

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<u>INTEGRATION BASICS:</u> DEFINITION OF INTEGRAL

- The integral is a fundamental concept in calculus.
- It represents the signed area under a curve.
- The definite integral $\int_a^b f(x)\,dx$ measures the accumulated quantity of $f(x)$ from a to b.
- In the indefinite integral, $\int f(x)\,dx$, the antiderivative of $f(x)$ is sought.
- Definite integrals are denoted as $\int_a^b f(x)\,dx$.

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<u>INTEGRATION BASICS:</u> RULES OF INTEGRATION

- Power Rule: $\int x^n \, dx = \frac{1}{n+1} x^{n+1}$, where $n \neq -1$.
- Substitution: u-substitution helps simplify integrals by making a substitution.
- Integration by Parts:
  $$\int uv = u \int v - \int (u' \int v)$$

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INTEGRATION BASICS: EXAMPLE PROBLEMS - POWER RULE

Evaluate $\int 3x^2\,dx$ using the power rule.
Using the power rule, $\int 3x^2\,dx = x^3 + C$, where C is the constant of integration.

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<u>INTEGRATION BASICS:</u> SUBSTITUTION & INTEGRATION BY PARTS

Evaluate $\int xe^x \, dx$ using substitution.
Let $u = x$, then $dv = e^x \, dx$. Applying integration by parts, we get
$\int xe^x \, dx = xe^x - \int e^x \, dx$.

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DERIVATIVES OF COMMON FUNCTIONS

- Polynomials: The derivative of $f(x) = ax^n$ is $f'(x) = nax^{(n-1)}$
- Exponentials: The derivative of $f(x) = e^x$ is $f'(x) = e^x$
- Logarithms: The derivative of $f(x) = \ln(x)$ is $f'(x) = \frac{1}{x}$

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TRIGONOMETRIC FUNCTIONS AND THEIR DERIVATIVES

- Sine function: The derivative of $f(x) = \sin(x)$ is $f'(x) = \cos(x)$
- Cosine function: The derivative of $f(x) = \cos(x)$ is $f'(x) = -\sin(x)$
- Tangent function: The derivative of $f(x) = \tan(x)$ is $f'(x) = \sec^2(x)$

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HYPERBOLIC FUNCTIONS AND THEIR DERIVATIVES

- Hyperbolic sine function: The derivative of $f(x) = \sinh(x)$ is $f'(x) = \cosh(x)$
- Hyperbolic cosine function: The derivative of $f(x) = \cosh(x)$ is $f'(x) = \sinh(x)$
- Hyperbolic tangent function: The derivative of $f(x) = \tanh(x)$ is $f'(x) = sech^2(x)$

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INTEGRALS OF ELEMENTARY FUNCTIONS

- The integral of a constant: $\int k\,dx = kx + C$
- Power rule: $\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C$
- Sum and difference rules: $\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx$

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BASIC TRIGONOMETRIC INTEGRALS

- $\int \sin(x)\,dx = -\cos(x) + C$
- $\cos(x)\,dx = \sin(x) + C$
- $\int \ln(x)\,dx = x\ln(x) - x + C$
- $\int e^x\,dx = e^x + C$

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APPLICATIONS

- Differentiation and integration are powerful tools in mathematics.
- They have wide applications in various real-world scenarios.
- Understanding these concepts is crucial for solving problems in optimization, physics, economics, and more.

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CONCLUSION:

Importance of Mastering Differentiation & Integration

- Foundation for Advanced Math
  - Calculus as a Gateway
  - Higher-level Mathematics
- Practical Applications
  - Problem-solving in Various Disciplines
  - Engineering, Physics, Computer Science
- Critical Thinking & Analytical Skills
  - Developing a Mathematical Mindset
  - Enhancing Problem-solving Skills

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Your attendance for this lecture can not be marked at this moment because it was scheduled for Tue., Nov. 26, 2024 8:00 AM (WAT) - Fri., Nov. 29, 2024 6:00 PM (WAT) and you are outside the specified lecture period.

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